P510/2
PHYSICS
Paper 2
July - August 2026
3 hours

KAMPALA INTEGRATED SECONDARY SCHOOLS EXAMINATIONS BUREAU
KAMSSA

KAMSSA JOINT MOCK EXAMINATIONS

Uganda Advanced Certificate of Education

PHYSICS

Paper 2

3 hours

Instructions to candidates

- *This examination paper consists of two compulsory items.*
- *Use blue or black ink. Any work done in pencil except graphs and diagrams will not be scored.*
- *Scores are awarded mainly for a clear record of the observations*
- *actually made, for their suitability, accuracy and for the use made of them.*
- *Details on the question paper should not be repeated in the responses.*
- *Graph paper is provided.*
  Silent non-programmable scientific calculators may be used.

© 2026 Kampala Integrated Secondary Schools Examinations Bureau *Turn Over*

CamScanner

ITEM 1

A regional physics outreach program organized by the Ministry of Education and Sports is working with secondary schools to improve experimental understanding of fundamental constants. One key objective is to determine the acceleration due to gravity using simple laboratory apparatus available in most schools.

Two independent experimental methods are proposed:

- A simple pendulum system
- A helical spring-mass system

Students are required to compare both methods and verify whether they produce consistent values ranging from 9.0 - 11.0 ms^{-2}.

In addition, the organizers explain that the accurately determined value of the constant will be used in a follow-up practical investigation to determine weight of a metre rule, which will later be used in balance experiments and torque measurements.

Task:

As a Physics student, you are required to conduct a **scientific investigation** to support the ministry in its move.

ITEM 2

A rural health center in Uganda is experiencing difficulties in reading small print on medicine labels and patient prescription forms, especially during evening hours when lighting is poor. To improve visual clarity for staff, the center acquires a device to be used as a simple magnifying aid.

However, after installation, the staff notice that the device does not always produce a sharp image at different working distances. The clarity of the magnified image varies, making it difficult to reliably read fine print and increasing the risk of medication errors.

To solve this problem, a group of physics students is invited to determine the optical property of this device, and to evaluate whether it is suitable for medical use. The device is considered suitable if its power is of magnitude in the range 8 - 12D.

Task

As a learner of physics, follow the instructions provided and use the result obtained to determine the suitability of the device.

METHOD I

(a) Mount the lens provided and place it facing the window.

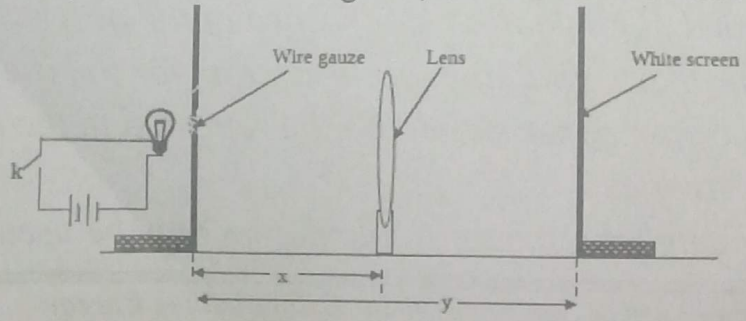
(b) Place the screen behind the lens and adjust it until a clear image of the distant object is obtained.

(c) Measure and record the distance, f_1, between the lens and the screen.

METHOD II

(a) Connect a pair of dry cells, a torch bulb and a switch K in series.

(b) Arrange the bulb, screen with a wire gauze, lens and the white screen as shown below.



CamScanner

(c) Adjust the position of the lens so that the distance *x*, *between* the wire gauze and the lens is 20.0cm.
(d) Close switch K
(e) Adjust the position of the white screen until a clear image of the wire gauze is obtained on the screen.
(f) Measure and record the distance, y, between the wire gauze and the white screen
(g) Open switch, K.
(h) Repeat procedures (c) to (g) for values of x = 25.0,30.0,35.0,40.0 and 45.0cm
(i) Record your results in a suitable table including values of $\frac{x^2}{y}$
(j) Plot a graph of *x* against $\frac{x^2}{y}$.
(k) Determine the slope, S of the graph.
(l) Find the intercept, C on the *x*-axis.
(m) Calculate f_2 using the expression $f_2 = CS$
(n) Calculate the power, P of the lens using an appropriate expression containing f_1 and f_2.
(o) Based on your findings, determine the suitability of the device.

END

CamScanner